FUNCTIONAL & TECHNICAL SPECIFICATION

Multi-Tier Inventory Architecture, Procurement Governance, Payment Ledger, & Lead Pipeline Dashboard

Author/Owner:	Nightvision Headquarters (HQ) Management
Target Audience:	Software Engineering Team / System Architects / QA Leads
Core Modules:	Inventory Mapping, POS Engine, Territorial AMC Routing, Accounts Receivable Trackers, Multi-Signature Procurement Workflows, Lead Management Visibility

1. System Scope & Structural Hierarchy

This functional blueprint details the logical design, data dictionary structures, and automated process guidelines required to build the unified corporate enterprise management platform. The application must strictly enforce the following network infrastructure hierarchy:

- **Headquarters (HQ):** Root administrative node. Maintains global view access of all operational sub-entities, approves corporate purchasing flows, and evaluates consolidated national leads.
- **Category A (Super-Distributors):** Regional master nodes. One distinct node operates per province across exactly **7 Provinces** (7 total nodes). Responsible for supplying regional fulfillment to Tier-B dealers and handling localized accounts receivable tracking.
- **Category B (Authorized Dealers):** Client-facing nodes. Exactly **7 Authorized Dealers mapped to each Super-Distributor** (Total: 49 nodes nationally). Handles point-of-sale executions, physical field collection, and area Project/AMC fulfillment.

2. Data Dictionary & Relational Schema Mapping

The engineering team must structure the database tables to retain referential integrity while tracking assets across dynamic states and financial stages.

Table 1: Unified Multi-Locator Inventory Stock (`inventory_ledger`)

Tracks current inventory units. Stock values must never be stored as simple bulk aggregates; they must be distinct rows mapped to explicit locations and specific physical sub-bins.

Field Technical Name	Data Type	Index Type	Business Integrity Constraints
`stock_record_id`	VARCHAR(36)	PRIMARY KEY	Unique entry UUID.
`sku_id`	VARCHAR(50)	FOREIGN KEY	References Product Master Table catalog SKU identifier.
`node_id`	VARCHAR(36)	FOREIGN KEY	References Master Org Nodes. Identifies current Category A or B custodian.
`locator_bin_status`	ENUM	NOT NULL	Enforce exact values: `'Fresh'` , `'Refurbished'` , `'Repairable'` , `'Damaged'` .
`quantity_on_hand`	INT	NOT NULL	Current count. Database constraints must prevent values < 0.
`last_updated_ts`	TIMESTAMP	NOT NULL	Automated system modification tracker timestamp.

Table 2: Dual-Stage Accounts Receivable Ledger (`payment_tracking_ledger`)

Manages downstream invoice collection tracking streams seamlessly.

Field Technical Name	Data Type	Null Matrix	Business Integrity Constraints
`payment_record_id`	VARCHAR(36)	PRIMARY KEY	Unique tracking record UUID.
`source_invoice_id`	VARCHAR(50)	NOT NULL	Links to the originating Demand Supply Order invoice request.
`payer_node_id`	VARCHAR(36)	FOREIGN KEY	The paying party (e.g., Cat B Dealer paying Cat A, or Cat A paying HQ).
`payee_node_id`	VARCHAR(36)	FOREIGN KEY	The receiving collector party (e.g., Cat A Super-Distributor or HQ).
`total_amount_due`	DECIMAL(15,2)	NOT NULL	Total invoice value baseline.
`total_amount_received`	DECIMAL(15,2)	DEFAULT 0.00	Accumulated verified transaction receipts.
`reconciliation_status`	ENUM	NOT NULL	Values: `'Unpaid'` , `'Partially_Paid'` , `'Settled'` , `'Disputed'` .

Table 3: Lead Pipeline Tracking Engine (`lead_pipeline`)

Captures client leads generated at any level, ensuring unified visibility for Nightvision HQ.

Field Technical Name	Data Type	Null Matrix	Business Integrity Constraints
`lead_id`	VARCHAR(36)	PRIMARY KEY	System generated lead tracking key code.
`province_id`	INT	NOT NULL	Enforces value alignment between `1` and `7`.
`target_dealer_id`	VARCHAR(36)	FOREIGN KEY	Points to assigned Category B node within the matched province boundary
`estimated_deal_value`	DECIMAL(12,2)	NOT NULL	Forecasted structural financial value for corporate pipeline planning.
`pipeline_stage`	ENUM	NOT NULL	Values: `'Identified'`, `'Proposal'`, `'Converted_To_Project'`, `'` `'Dead'`.

3. Dual-Track Corporate Procurement Workflows

The procurement modules must strictly enforce validation paths before any Purchase Order (PO) can be initialized. The backend must block manual process overrides.

Workflow A: Local Procurement Chain

Applies to purchasing sourced domestically. The digital requisition must navigate this exact horizontal linear path:

Stage 1: Entry	Stage 2: Check 1	Stage 3: Check 2	Stage 4: Final Gatekeeper
Requisition Drafted by Operations Team	**COO Approves / Rejects** (Digital Signature)	**CEO Approves / Rejects** (Digital Signature)	**Head of HR Operations Signs Off & Releases PO**

Workflow B: International Procurement Chain

Applies to global sourcing requests. This track introduces technical architecture scrutiny before corporate leadership sign-off:

Stage 1: Entry	Stage 2: Technical	Stage 3: Corporate 1	Stage 4: Corporate 2	Stage 5: Gatekeeper
Requisition Drafted	**CTO Evaluates & Signs**	**CEO Evaluates & Signs**	**COO Evaluates & Signs**	**Head of HR Operations Signs Off & Releases PO**

> **Technical Rule for Engineers: State Machine Implementation**
>
> Implement the procurement routing engine as a deterministic state machine. The database column `approval_status` must not jump order. For international orders, Stage 5 cannot change to 'Approved' unless Stages 2, 3, and 4 verify legitimate cryptographically secure user approval hashes.

4. System Business Logic Rules

A. Automated Point of Sale (POS) Inventory Deductions

- Every single customer transaction running at a Category B Authorized Dealer register must transmit payload packets to the centralized inventory tracking engine.
- The backend must query the dealer's specific `inventory_ledger` record matching the item SKU. It must only decrement counts located inside the `'Fresh'` or `'Refurbished'` database status bins. It must explicitly block transactions if stock is restricted to `'Repairable'` or `'Damaged'` states.

B. Regional Project & AMC Allocation Engine

- When a contract or operational ticket is ingested by the platform, it is tagged with a distinct regional target `province_id`.
- The system allocation interface must automatically restrict dealer assignments based on geographic location. Users must only see the 7 Category B dealers registered underneath that specific province's Category A Super-Distributor. Cross-boundary assignments must fail system validation rules.

C. Hierarchical Supply Chain Demand Module

- **Dealer Pull:** When Category B dealer stocks drop below minimum safe limits, a replenishment request drops into the Category A regional fulfillment hub panel.
- **Super-Distributor Consolidation:** Category A can instantly fulfill requests from local safety stock, or generate an upstream bulk stock requisition routing directly up to Headquarters.

5. Developer Endpoint Reference Blueprint

Provide the following JSON structure model to frontend engineers creating the procurement request transmission functions:


```
{
  "procurement_request_meta": {
    "origin_scope": "INTERNATIONAL",
    "initiator_user_id": "usr_hq_ops_771",
    "total_estimated_value_usd": 145000.00
  },
  "requested_line_items": [
    {
      "sku_id": "NV-GOGGLE-GEN3-PRO",
      "quantity_ordered": 250,
      "unit_cost_usd": 580.00
    }
  ],
  "approval_routing_stack": [
    { "role": "CTO", "execution_sequence": 1, "signed_status": "APPROVED", "timestamp": "2026-05-26T04:12:00Z" },
    { "role": "CEO", "execution_sequence": 2, "signed_status": "PENDING", "timestamp": null },
    { "role": "COO", "execution_sequence": 3, "signed_status": "QUEUE", "timestamp": null },
    { "role": "HEAD_HR_OPS", "execution_sequence": 4, "signed_status": "QUEUE", "timestamp": null }
  ]
}
```